ShelfSense

AI-Based Retail Merchandising Audit: progress summary (Sprints 1-3)

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1. Problem and goal

Beverage and FMCG companies (e.g. Coca-Cola, PepsiCo/Transcom, Akij) pay shops for merchandising: they **supply branded fridges** on condition that only their products are stored inside, and **pay extra for prime placement** (front, eye level). To check this, they send field audit representatives. These audits are expensive, and representatives are often **bribed by shopkeepers and fabricate their reports**.

ShelfSense replaces the representative's word with **verifiable video evidence**: a walkthrough video of the shelves is analysed automatically, and every finding points to a frame a human can check. The goal is to **reduce audit cost for companies** and keep shopkeepers accountable.

Audit KPIs

KPI	Question answered
Cooler purity	Does the company's fridge contain only the company's products?
Paid placement	Is the brand on the agreed shelf / at the front, as paid for?
Out-of-stock	Are there empty shelf gaps where products should be?
Share of shelf	What share of the facings belongs to the client brand vs competitors?
Price tag / POSM	Are price labels and promotional material present and visible?

Design principle: a false accusation ("competitor found in your fridge") harms an honest shopkeeper, so ShelfSense favours **precision over recall** for violations, attaches evidence to every finding, and marks uncertain cases for human review.

2. Approach: hybrid detector + VLM pipeline

Vision-language models (VLMs) are weak at counting densely packed products, so detection and reasoning are split into stages. All heavy models are small enough for free cloud GPUs (Kaggle); the rest runs on a CPU-only laptop (8 GB RAM).

Stage	What it does	Status
1. Extraction	Sample frames, drop blurry and near-duplicate frames	Done (Sprint 2)
2. Detection	YOLO nano finds every product facing and empty gap	Feasibility tested
3. Brand / VLM	Identify the brand of each detected product; read tags	Baseline done (Sprint 3)
4. Rules	Compare with fridge ownership, placement agreements, KPIs	Planned (Sprint 5)
5. Report	Labelled video (owner = green, competitor = red, unsure = yellow), JSON report, evidence frames, web demo	Planned (Sprints 5-6)

3. Sprint 1: setup and problem definition

- GitHub repository with a modular layout (`src/extraction, detection, vlm, rules, report`), central configuration file, and an environment check script.

- Environment: Python 3.14, OpenCV, CPU PyTorch, Ultralytics; Kaggle (free T4 GPU) for model inference.
- Scope fixed to three categories (beverages; soaps and detergents; biscuits and snacks) and five KPIs.

4. Data collected

Three real phone videos of branded beverage fridges recorded in shops in Bangladesh.

Video	Content	Length	Resolution
fridge_01	Coca-Cola (CCI) fridge: Sprite, Fanta, Coca-Cola; a table fan blocks part of it	2.3 s	576 x 1024
fridge_02	Mountain Dew (Transcom) fridge, mostly empty, only Aquafina water	6.4 s	576 x 1024
fridge_03	Two Mojo fridges with mixed brands, wide shot	2.4 s	576 x 1024



Figure 1. One frame from each collected video.

Limitation: the clips are short (11 s in total) and were compressed by WhatsApp. They are enough to test the pipeline and find failure patterns, not to report final statistics. Recording more full-quality footage is the first target for next week.

5. Sprint 2: keyframe extraction

A walkthrough video contains many blurry and repeated frames. The extractor reduces it to a small set of sharp, distinct keyframes, processing one frame at a time so it fits in 8 GB of RAM.

- **Sampling by timestamp** (not frame number), because phone videos have a variable frame rate.
- **Resize** wide frames to at most 1280 px (never upscale).
- **Blur filter:** drop frames whose blur score is above a threshold.
- **Duplicate filter:** drop frames whose perceptual hash (pHash) is within 6 bits of the last kept frame.
- Every decision is logged (`frames_log.csv`) with a summary (`stats.json`); covered by automated tests.

Results (sampling at 4 frames per second)

Video	Sampled	Kept	Blurry	Duplicate
fridge_01	9	9	0	0
fridge_02	26	16	0	10
fridge_03	10	5	0	5
Total	**45**	**30**	**0**	**15**

Key finding: choosing the blur metric

The standard sharpness score (variance of the Laplacian) **failed on real fridge footage**: it depends on scene content, so sharp close-ups scored low and were thrown away, while blurred frames of busy scenes scored high. We replaced it with the no-reference **blur effect metric of Crete et al. (2007)**, which compares each frame with a re-blurred copy of itself.

Metric	Sharp real frames	Same frames, artificially blurred	Usable threshold?
Laplacian variance	49 - 1031	3 - 147	No (ranges overlap)
Crete blur effect	0.24 - 0.39	0.56 - 0.72	Yes, 0.5



Figure 2. Four sharp fridge_02 frames scored by both metrics. The Laplacian filter (threshold 100) would drop the close-ups, the most useful frames in the clip; the Crete metric (threshold 0.5) rates all four as sharp.

6. Sprint 3: zero-shot VLM baseline

Setup. Qwen2-VL-2B-Instruct (2 billion parameters), no training, run on a Kaggle T4 GPU. Five audit questions with JSON answers (fridge brand, brands inside, counts, empty space, price tags), asked in two modes: **image mode** (each of the 30 keyframes) and **video mode** (up to 8 keyframes per clip given together as a video). 165 answers per run, scored automatically against hand-made ground truth. Two prompt versions: v1 (baseline) and v2 (rewritten to target the v1 failures; model and settings unchanged).

Metric (mean over videos)	v1 image	v1 video	v2 image	v2 video
Unreadable answers (total)	22	2	8	0
Seconds per answer	3.6	3.5	2.9	2.6
Fridge brand correct	0.67	0.33	0.67	0.67
Brand precision	0.27	0.30	0.13	0.25
Brand recall	1.00	0.50	1.00	1.00
Empty space correct	0.33	0.67	0.67	0.67
Price tags correct	0.00	0.00	0.33	0.00
Competitor (purity) correct	0.00	0.00	0.00	0.00

Scores are 0-1 (1 = always correct). Three videos only: a change of 0.33 means one video changed its answer, so the table shows failure patterns rather than statistically significant differences.

Failure cases observed

Failure	Example from our data
Invented brands	Listed Pepsi, 7up, Dr Pepper, Lipton in a Coca-Cola fridge; Pepsi and Sprite in a fridge holding only water
No real counting	Answered "10" for every brand (or "1" / "0" for every brand)
Invented price tags	Sequences such as 1, 2, 3, 4 ... and TK 10, TK 15, TK 20 ...; copied the "<price>" placeholder from the prompt
Repetition loops	"Fanta", "Fanta", "Fanta" ... until the answer was cut off (unreadable JSON)
Reading the wrong text	"nojo" for Mojo; shop sign ("Kazfarm Kitchen") or owner text ("Transcom Beverages Limited") given as the fridge brand
Limits become targets (v2)	"At most 8 brands" produced exactly 8 brands; "at most 5 prices" produced 5 invented prices

Conclusion. Better prompts fixed **format** problems (unreadable answers 24 to 8, faster answers, specific locations) but not **perception** problems (invented brands, counting, price tags). Because invented competitor brands would falsely accuse honest shopkeepers, a VLM looking at the whole fridge **cannot be used as audit evidence**. This supports the hybrid design: detect each product first, then classify one product at a time.

7. Detection feasibility test (preview of Sprint 4)

A pretrained YOLO11n model (COCO "bottle" class, no training) was run on our keyframes on the laptop CPU (about 70-80 ms per frame). It finds most bottles in a close-up, but misses the green bottles behind glare, misses all cans and most products in the wide shot, and detects bottles standing on top of the fridge. **Fine-tuning on fridge images is therefore required.**



fridge_01: close-up fridge_03: wide shot

Figure 3. Pretrained YOLO11n detections (green boxes) with no fine-tuning.

8. Target output

An annotated video for the company's review system: every detected product is boxed and tracked; the fridge owner's products are **green**, competitor products **red**, and uncertain products **yellow** (human review), with a purity score and time-stamped evidence frames for every violation. Brand identification will match each product crop against reference examples of each brand, which is fast on a CPU and can answer "not sure" instead of guessing.

9. Next steps

- Record 10-15 close-up, full-quality fridge videos, including fridges with mixed brands (real purity violations) and cans.
- Verify and complete the ground truth (brands of fridge_03, product counts).
- Annotate 200-300 frames and fine-tune YOLO11n on Kaggle; report mAP.
- Build a reference set of brand crops and test per-product brand identification.
- Prototype the labelled green/red/yellow video; then integrity checks (fridge identity, re-used footage via perceptual hashes, file hashes) and the web demo.

Reference

F. Crete, T. Dolmiere, P. Ladret, M. Nicolas, "The blur effect: perception and estimation with a new no-reference perceptual blur metric," *Proc. SPIE Human Vision and Electronic Imaging XII*, 2007.